

FLOW DIAGRAM EXPLAINER

HFB Proactive AI Chatbot

How a behavioural signal on the portal becomes a completed booking — the end-to-end flow, and a plain-language guide to every step in it.



How to read this document

Each of the five stages above has its own page: a detailed flowchart at the top, and a one-line explanation of every box in that flowchart underneath. Follow a customer top to bottom through Sections 1→5, or jump straight to the stage you're implementing.

 Customer-facing

 Backend / system

 Decision point

 End state

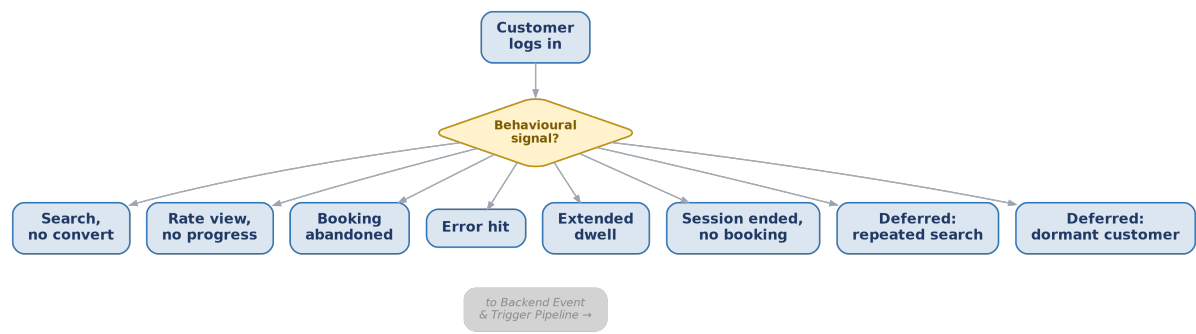
Contents

Five stages, followed in order, trace one customer end to end — from the first behavioural signal on the portal through to a booked conversion or a human handoff.

1**Customer Journey***What the customer does on the post-login HFB portal that can start the flow.***p.3****2****Backend Event & Trigger Pipeline***How a raw behavioural signal — or a human-handoff request — becomes an approved, routed engagement, in Python.***p.4****3****Conversation Orchestrator***How the bot decides what to say, and how it stays truthful.***p.5****4****Customer Response***What happens after the customer actually sees the message.***p.6****5****Admin, Audit & Reporting***How every outcome feeds back into visibility and control.***p.7**

1 Customer Journey

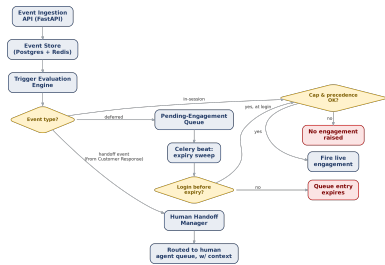
What the customer does on the post-login HFB portal that can start the flow.



A	Customer logs in The customer authenticates into the HFB portal, so every behaviour that follows is already tied to a known identity.
B	Behavioural signal detected The portal or booking widget emits an event the moment the customer's behaviour matches a trackable pattern.
C	Search, no convert Customer ran a search on the booking widget but did not complete a booking.
D	Rate view, no progress Customer viewed rates or vehicle options but did not move on to the booking step.
E	Booking abandoned Customer started the booking flow and exited partway through; the exact step they left at is captured.
F	Error hit Customer hit an error or validation failure while searching or booking.
G	Extended dwell Customer spent an unusually long time on a single step without progressing.
H	Session ended, no booking Customer logged in and left the portal without making a booking.
I	Deferred: repeated search Customer searched the same route or dates across more than one session without ever booking.
J	Deferred: dormant customer A previously active customer has not booked for a defined dormancy period.

2 Backend Event & Trigger Pipeline

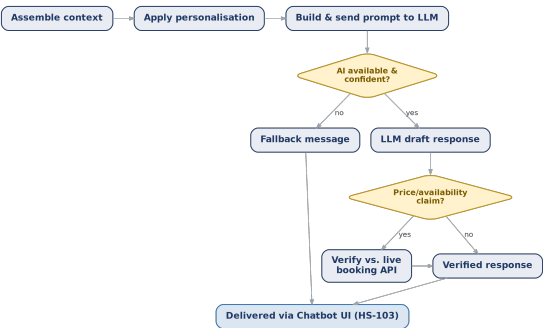
How a raw behavioural signal — or a human-handoff request — becomes an approved, routed engagement, in Python.



K	Event Ingestion API A FastAPI service receives, validates and normalises every behavioural event coming from the portal and widget.
L	Event Store Events are written to Postgres and a Redis stream, so the pipeline has durable storage and near-real-time delivery.
M	Trigger Evaluation Engine Evaluates every incoming event, and any scheduled checks, against the rules an admin has configured — including handoff requests (see below).
N	Event type decision The engine decides whether a match should engage the customer live now, be queued as a deferred trigger, or is a handoff event from the bot itself.
O	Frequency cap & precedence check Confirms the customer hasn't been re-engaged by this trigger too recently, and resolves which trigger wins if several match at once.
P	Pending-Engagement Queue Deferred matches are stored here to be raised the next time the customer logs in, rather than acted on immediately.
Q	Fire live engagement An approved trigger is handed off to the Conversation Orchestrator to open a conversation.
R	Celery expiry sweep A scheduled job walks the queue and expires entries once they're too old to still be relevant.
S	Login-before-expiry check At each login, the system checks whether a queued deferred engagement is still valid or has already expired.
HM	Human Handoff Manager Plan change: handoff is now handled here, not ad hoc in the conversation flow. Receives the handoff event and applies the same admin-configured routing rules (skill, team, language, priority).
Z3	Routed to human agent queue Dispatches the handoff, with the full conversation context attached, into the existing support queue for an agent to pick up.
Z1	No engagement raised The match failed the frequency-cap or precedence check, so nothing is sent to the customer this time.
Z2	Queue entry expires The deferred engagement aged out before the customer returned, so it's discarded unraised.

3 Conversation Orchestrator

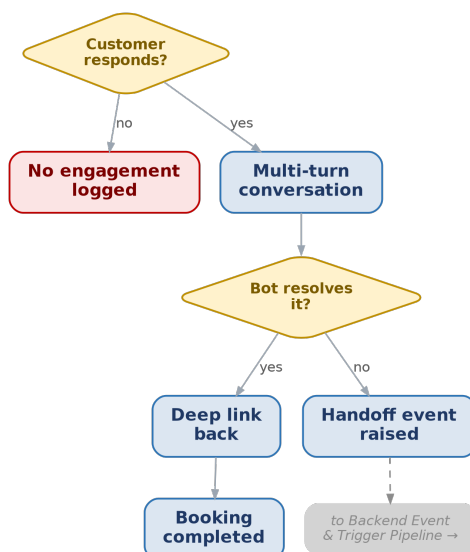
How the bot decides what to say, and how it stays truthful.



T	Assemble context Combines the firing trigger with the customer's profile and booking history, so the bot knows exactly what just happened.
U	Apply personalisation Adjusts tone and content for the customer's Customer Type, Market/Region and language before anything is generated.
V	Build & send prompt The assembled context is turned into a prompt and sent to the configured LLM provider.
W	AI availability / confidence check Confirms the AI service responded in time and with enough confidence for its answer to be used.
X	Fallback message If the AI is unavailable or unsure, a safe templated message is used instead of a generated one.
Y	LLM draft response The model generates a draft reply grounded in the assembled context.
AA	Price / availability check Flags whether the draft response makes any factual claim about a price, rate or availability.
AB	Live verification Any such claim is checked against the real booking system before it's allowed to reach the customer.
AC	Verified response The response, verified wherever it needed to be, is finalised for delivery.
AD	Delivered via Chatbot UI The message reaches the customer through the existing HS-103 chat interface, in the context they were already in.

4 Customer Response

What happens after the customer actually sees the message.



AE

Customer responds?

Checks whether the customer engages with the proactive message at all.

AF

No engagement logged

If the customer never responds, that's recorded so it counts correctly against the engagement-rate metric.

AG

Multi-turn conversation

If they do respond, the bot continues the conversation with full context retained across turns.

AH

Bot resolves?

Determines whether the bot can actually help with what the customer needs, or should hand off.

AI

Deep link back

If resolved, the bot links the customer straight back to their prior search or the booking step they left.

AJ

Booking completed

The customer finishes the booking, ideally as a direct result of the bot's intervention.

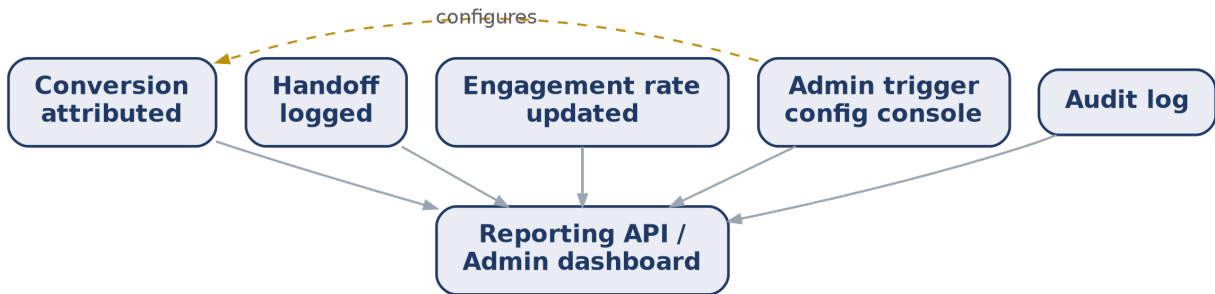
AK

Handoff event raised

Plan change: if the bot can't help, it no longer handles the handoff itself — it raises an event that's routed and dispatched by the Human Handoff Manager in Section 2, with context preserved.

5 Admin, Audit & Reporting

How every outcome feeds back into visibility and control.



AL Conversion attributed
A completed booking that follows an intervention is logged against the trigger that started it.

AM Handoff logged
Every handoff the Human Handoff Manager routes (Section 2) is recorded here, feeding the handoff-rate metric.

AN Engagement rate updated
Every response, or lack of one, feeds the engagement-rate metric for its trigger.

AO Reporting dashboard
All of the above rolls up into admin-facing reporting inside the portal.

AP Admin trigger config
Admins create, enable, disable and tune triggers here, with no code change required, feeding straight back into the Trigger Evaluation Engine.

AQ Audit log
Every configuration change is recorded for compliance and traceability.